Joy Nkatha

BASIC ELECTRONICS

ISCED UNIT CODE: 0714 441 04A
TVET CDACC UNIT CODE: IT/CU/ICTA/CC/01/5/MA

CONTINUOUS ASSESSMENT TEST (CAT 2)

85

TIME: 2 HOURS
TOTAL MARKS: 70

INSTRUCTIONS TO THE CANDIDATE

i. *Answer all questions.*

ii. *Write your answers clearly and legibly.*

iii. *Draw well-labelled diagrams where required and show all working for calculations.*

1. A technician is examining electronic components for use in a control circuit.

a) Define semiconductor material. (2 marks)

b) Describe the structure of a silicon atom, indicating the number of valence electrons. (3 marks)

c) Distinguish between conductors and semiconductors based on the energy band. (4 marks)

2. A trainee is preparing semiconductor materials for the manufacture of electronic components.

a) Describe how doping produces P-type semiconductor material. (3 marks)

b) Describe how doping produces N-type semiconductor material. (3 marks)

c) State two differences between P-type and N-type semiconductor materials. (4 marks)

3. A technician connects a PN-junction diode to a DC power supply to determine its operating characteristics.

a) Draw a labelled diagram of a PN-junction diode under forward bias. (4 marks)

b) Explain how forward bias affects the depletion region and current flow in the diode. (4 marks)

c) Draw a labelled diagram of a PN-junction diode under reverse bias. (4 marks)

d) Explain how reverse bias affects the depletion region and current flow in the diode. (4 marks)

4. Using the theorems of Boolean algebra , prove the following identities: (6marks)

(i) $(A+\overline{B})\cdot C+(B+\overline{C})\cdot A+\overline{A}\cdot B=A+B+C$

(ii) $\overline{X}\cdot(X+Z)+\overline{Y}+Y\cdot Z=\overline{Y}+Z$

5. Simplify the following functions of Boolean algebra using the K

i) $F(a,b,c,d) = \sum (0,1,2,8,9,10)$

ii) $F(a,b,c,d) = \sum (0,1,4,6,9,13,14,15)$

6. A computer system uses different storage technologies to store data and instructions.

a) Classify each of the following as semiconductor, magnetic or optical memory:

i. Solid-state drive (SSD) (1 mark)

ii. Hard disk drive (HDD) (1 mark)

iii. DVD (1 mark)

b) Distinguish between internal storage and online storage. (2 marks)

c) State two differences between main memory and secondary storage. (2 marks)

7. An engineer is designing a digital control circuit that receives two inputs, A and B.

a) Draw the symbols for the AND, OR and NOT gates. (3 marks)

b) Draw the truth table for an XOR gate. (3 marks)

c) State one difference between NAND and NOR gates. (2 marks)

d) Simplify the following Boolean expression using Boolean algebra. (4 marks)

$$Y = (A \cdot B) + (A \cdot \overline{B})$$

(b) Silicon has 14 electrons in shells of 2, 8 and 4, has 4 valence electrons

(c) Conductors have no energy gap while semiconductors have a small gap of 1ev.
Conductors has very low resistance and semiconductors have moderate resistance.
Conductors resistance rises with temperature while semiconductor resistance falls.

2(a) P-type : Dope pure si with a trivalent impurity. Each impurity atom leaves one missing bond called hole(majority carriers)

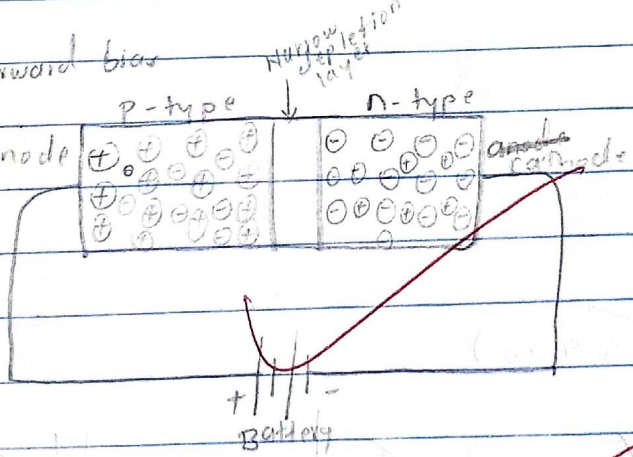
(b) N-type : Dope pure si with a pentavalent impurity. Each impurity atom gives one free electron. Electrons are majority carrie

(c) P type
Uses trivalent dopants
Holes are majority carriers

N-type
Uses pentavalent dopants.
Electrons are majority carriers.

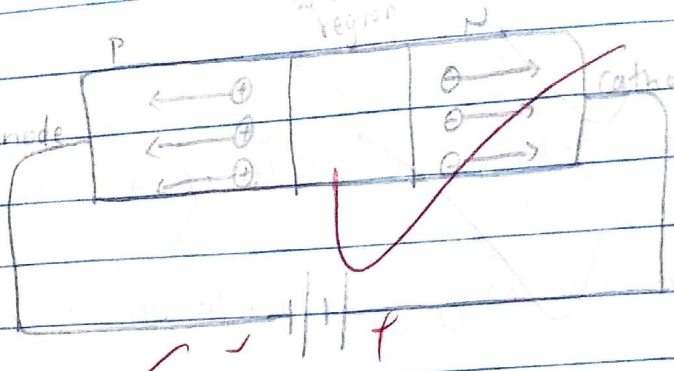
3 Forward bias

(a)



(b) Barrier potencial decreases
Resistance decreases
Current flows through the diode
Depletion region becomes narrow.

(c)



(d) Barrier potential increases
Resistance increases
Depletion region becomes wider
Only a very small leakage current flows.

4 (i)

$$(AC + \bar{B}C) + (AB + A\bar{C}) + \bar{A}B$$
$$= AC + \bar{B}C + AB + A\bar{C} + \bar{A}B$$
$$= A(C + \bar{C}) + B(A + \bar{A}) + \bar{B}C$$
$$= A + B + \bar{B}C$$
$$= A + B + C$$

(ii)

$$= \bar{X}X + \bar{X}Z + \bar{Y} + YZ$$
$$= 0 + \bar{X}Z + \bar{Y} + YZ$$
$$= \bar{Y} + Z(\bar{X} + Y)$$
$$= \bar{Y} + Z$$

5 (i) $F(a,b,c,d) = \Sigma(0, 2, 8, 9, 10)$

$$= F = \bar{b}\bar{c} + \bar{b}\bar{d}$$

(ii) $F(a,b,c,d) = \Sigma(0, 1, 4, 6, 9, 13, 14, 15)$

$$F = abc + a\bar{c}d + \bar{b}\bar{a}d + \bar{a}\bar{b}\bar{c}$$

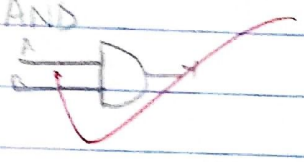
Show the working!

(i) SSD - Semiconductor storage
(ii) HDD - Magnetic storage
(iii) DVD - Optical storage.

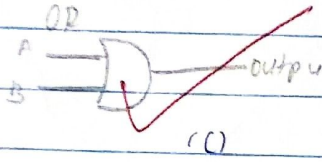
(b) Internal storage - found inside a computer
Online storage - stores data on internet servers.

(c) ~~Primary~~ Secondary storage.
- Main memory
- Volatile
- Fast and small
- non-volatile
- slower and large.

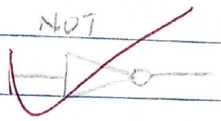
7 (a) AND



OR



NOT



(b)

A	B	Y
0	0	0
0	1	1
1	0	1
1	1	0

(c) NAND is the opposite of AND
NOR is the opposite of OR.

(d) $Y = (A \cdot B) + (A \cdot \bar{B})$

$Y = A(B + \bar{B})$

$Y = A(1)$

$Y = A$